

Expense Reimbursement Management System

Business Objective

Design and develop a web-based application that enables employees to:

Manage reimbursement requests

Track expense submissions

Monitor approval status

Calculate total claimed and approved amounts

The system should provide a centralized dashboard for reimbursement insights.

System Overview

This application allows an Employee to create and manage one or more Reimbursement Requests. Each request belongs to a specific Category.

The system performs financial tracking to determine:

Total requested amount

Total approved amount

Total pending amount

Reimbursement summary

Entities & Data Model

Employee

Represents the system user who submits reimbursement requests.

Reimbursement

Represents a financial claim submitted by an employee.

Category

Represents the type of expense (Travel, Food, Office, etc.).

Entity Associations

Employee → Reimbursement

One Employee can submit multiple Reimbursement

Requests. **Category → Reimbursement**

One Category can contain multiple Reimbursement

Requests.

Core Functional Requirements

Employee Management

Create employee

Update employee details

View employee list

Delete employee

Category Management

Create category

Update category

View categories

Delete category

Reimbursement Management

Create reimbursement request

Update reimbursement request

View reimbursements

Delete reimbursement

Financial Calculations

The system must automatically compute:

Total Requested Amount

Sum of all reimbursement amounts

Total Approved Amount

Sum of all approved reimbursements

Total Pending Amount

Requested Amount – Approved Amount

Reimbursement Summary

Total requested amount

Total approved amount

Total pending amount

Category-wise expense performance

Application Screens / Routes

Authentication (Optional Assumption)

/login

Dashboard

Route: /dashboard

Purpose:

Display total requested amount

Show approved amount summary

Show pending reimbursement summary

Employee Management

Route: /employees

Features:

List employees

Add employee

Edit employee

Delete employee

Category Management

Route: /categories

Features:

List categories

Create new category

Edit category

Delete category

Reimbursement Management

Route: /reimbursements

Features:

View reimbursements

Create reimbursement request

Edit reimbursement

Delete reimbursement

Technical Implementation Requirements

Backend (Spring Boot)

RESTful APIs

Layered architecture:

- Controller

- Service

- Repository

JPA relationships

Exception handling

Validation

Unit tests (JUnit + Mockito)

Code coverage recommended

Frontend (React)

React components

Redux for state management

React Router for navigation

Axios / Fetch for API calls

Basic styling (CSS or Bootstrap)

Database (MySQL)

Proper foreign key constraints

Normalized structure

Deployment

Frontend built using npm run build

Backend deployed on Apache Tomcat

Database connected via Spring Boot configuration

Assumptions

Approval workflow can be basic.

Currency is single (e.g., INR or USD).

No external finance API integration

required. Authentication can be basic or optional.